

Sound, space & *perception.*

Sound artist · Physicist · Researcher
Buenos Aires, Argentina

[Vimeo ↗](#)

[YouTube ↗](#)

[Website ↗](#)



Sonic Crystal Room · Manuel Eguía / Oscar Edelstein

Artistic practice and scientific research

My professional background is research in physics, nonlinear dynamics, neuroscience and acoustics. For 25 years, I have worked at an art school, training artists and connecting art and science through laboratory work and international collaborations. My artistic practice has developed through this work.

I develop works based on current questions in sound, acoustics and auditory perception. These projects examine how acoustic materials affect listening, how virtual environments alter the perception of space, and how musical practices relate to their acoustic surroundings.

Scientific models, experiments and technical development are part of this practice. The works take shape in relation to particular spaces, acoustic conditions and ways of listening.



Manuel Eguía is an artist and physicist based in Buenos Aires. He holds a PhD in Physics from the Universidad de Buenos Aires (2002), is a CONICET researcher and a full professor at the Escuela Universitaria de Artes, Universidad Nacional de Quilmes. He founded LAPSo, the Laboratory of Acoustics and Sound Perception, in 2007, and founded Proyecto GRAPa in 2017. Since 2026, he directs UNQ's Instituto de Investigaciones en Arte, Ciencia, Tecnología y Sociedad (ArCiTeS).

Sonic Crystal Room

A room that becomes an instrument

Manuel Eguía / Oscar Edelstein



Acoustic installation · Performance environment

[Watch the work ↗](#)

[Sonic Crystal Room website ↗](#)

[Project explanation · 5:57 ↗](#)

An architecture of mobile modules and rotating wooden columns transforms sound through physical acoustics. The room itself becomes a musical instrument.

In Sonic Crystal Room, we use the acoustic behaviour of a space as a material for composition and performance. The columns form a sonic crystal: a repeated structure that changes how sound travels. Reconfiguring the structure changes what a listener hears and where the sound seems to be.

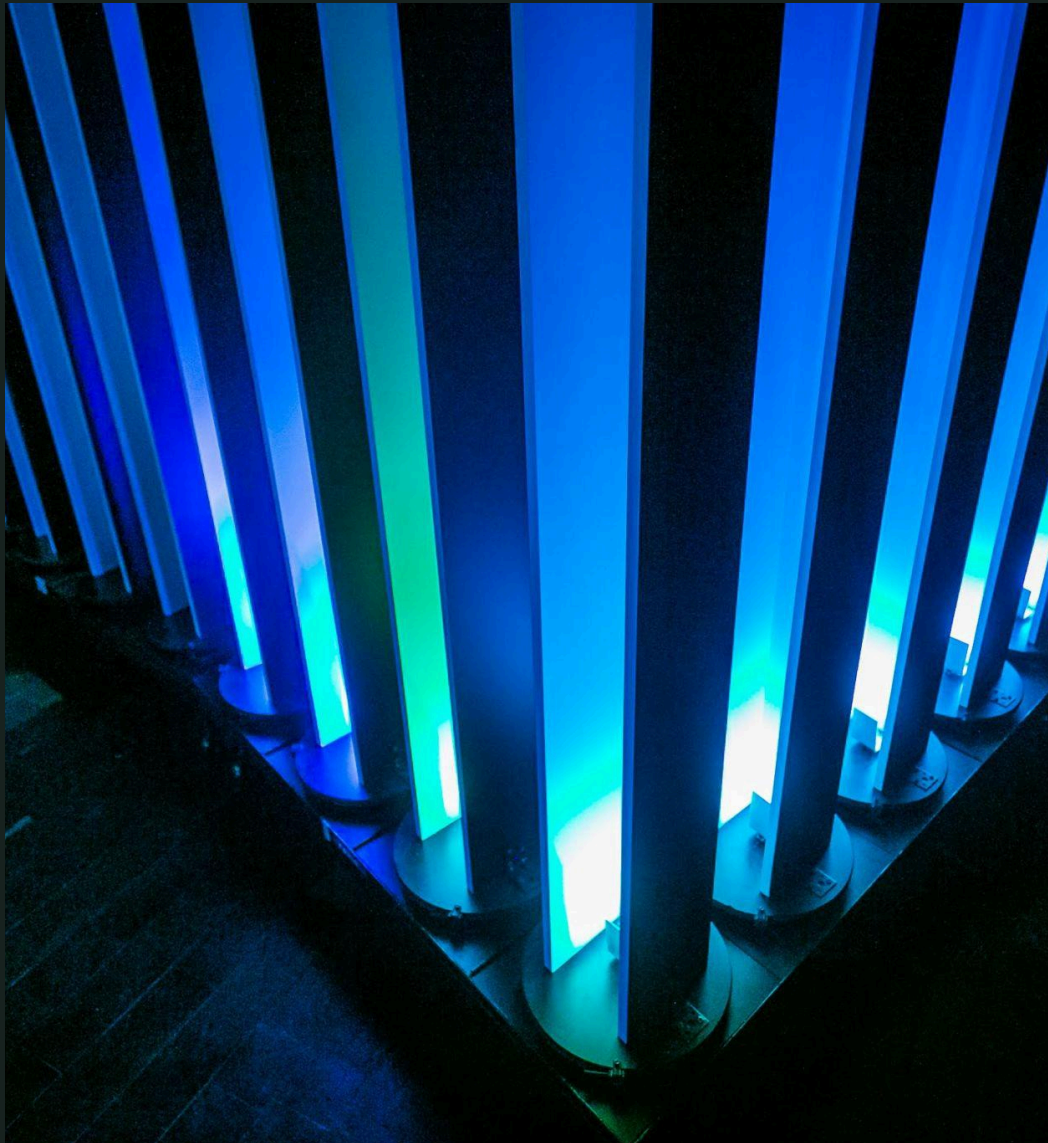
The work makes architecture an active part of a performance. Movement, apparent distance and shifts in acoustic perspective emerge from the relationship between sound sources, the columns and the listener.

ROLE

Co-creator with Oscar Edelstein; acoustic research and development.

CONTEXT

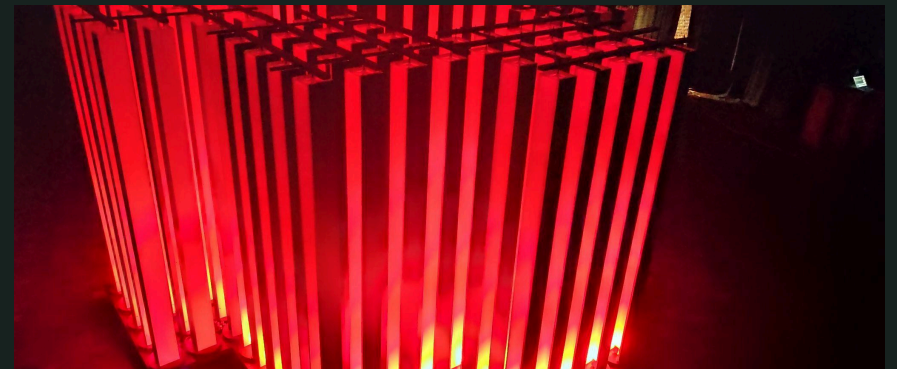
First presented at UNQ in 2019. Presented in the opening programme of the Festival de Nueva Ópera, cheLA, Buenos Aires, September 2024.



Composing with the space itself

We use mobile modules and rotating columns to change the room's acoustic properties during a performance. Focusing, redirection and scattering become compositional resources, alongside the sound produced by musicians or loudspeakers.

Outside performances, the installation can be explored as a three-dimensional score. Visitors move through an architecture whose geometry and material properties organize the listening experience.



Mobile structures, changing acoustic perspectives.
Manuel Eguía / Oscar Edelstein

The Wind Chimes

Tubular instruments for a bridge and a river

Buenos Aires Sonora / LAPSo



Site-specific instruments · Collective performance
Photographs: Ximena Martinez / Buenos Aires Sonora archive.



Five large tubular wind chimes, suspended from the Nicolás Avellaneda transporter bridge, were played during a collective sound performance over the Riachuelo.

With Buenos Aires Sonora and the LAPSo team, we built instruments for El Puente Suenan II [Invocaciones], presented within Sigismond de Vajay's Lluvia de Arañas sobre el Riachuelo.

We designed the instruments around resonating metal tubes and a suspended striking mechanism. Performers set them in motion from below, making the bridge, the river and the space between them part of the performance.

ROLE

Instrument design; construction and performance with LAPSo, in collaboration with Buenos Aires Sonora.

CONTEXT

El Puente Suenan II [Invocaciones], within Lluvia de Arañas sobre el Riachuelo. La Noche de los Museos, Puente Transbordador Nicolás Avellaneda, La Boca, Buenos Aires, 12 November 2011.

[Backstage: Lluvia de Arañas sobre el Riachuelo · 9:02 ↗](#)



From the laboratory to the river

We built and tested the tubular instruments at LAPSo before installing them on the bridge. The installation photographs show the instruments being assembled, suspended and activated from a boat below.

The wind chimes formed one part of Buenos Aires Sonora's performance. Other performers used the bridge's metal structure as a sound source, alongside recorded voices and live electronics.

Instrument concept: Hernán Kerlleñevich. Design: Manuel Eguía. Construction: Manuel Eguía, Ignacio Spiousas, Mauricio Biviano, Pablo Riera and Pablo Etchemendy. Performers on the large chimes: Manuel Eguía, Ignacio Spiousas, Mauricio Biviano, Pablo Riera and Ramiro Vergara.

[Backstage: Lluvia de Arañas sobre el Riachuelo · 9:02 ↗](#)

Photographs: Ximena Martinez / Buenos Aires Sonora archive.

IRIS

Rotating discs, changing sound

Leonardo Salzano / Manuel Eguía · LAPSo



Kinetic acoustic sculpture · Musical performance
Teaser image: Ezequiel Hilbert, 2019.

[IRIS teaser · 1:39 ↗](#)

[Related study · JNMR 2022 ↗](#)

A kinetic acoustic sculpture of two independently rotating, perforated discs. Their relative alignment changes how sound passes through the structure, affecting timbre and spatial projection.

Developed at LAPSo from Leonardo Salzano's original design, IRIS forms part of our work on acoustic materials for performance. We combine physical experimentation, construction and programming to make the geometry of the sculpture adjustable during a musical work.

Placed between a source and listeners, the discs alter the transmission of particular frequency bands. Rotation changes the relationship between the two layers, allowing the sculpture to modify the sound's spectral balance and apparent location.

ROLE

Scientific advice, acoustic research, construction and programming with Leonardo Salzano. Original and technical design: Leonardo Salzano.

CONTEXT

2018–2019 version documented in the teaser. Presented in the Sonic Crystal Room programme at UNQ in April 2019. Related study published in *Journal of New Music Research* (2022).

Transit

Inside, outside, elsewhere

Manuel Eguía / Mauro Zannoli



Site-specific virtual reality installation

[Watch the virtual journey ↗](#)

[Northern Sustainable Futures ↗](#)

A virtual journey begins in an exact replica of the room the participant occupies. Familiar space gradually opens into impossible interiors and transformed landscapes.

We conceived Transit as a series of site-specific virtual reality experiences. Each episode starts from the participant's immediate physical surroundings, making the transition into a virtual world difficult to locate. Local architecture, landscape and cultural references become the material of the work.

The first episode takes place in iM Konsthall, a mobile gallery built inside a shipping container. Its interior multiplies and transforms through a sequence of spatial folds, until the participant finds themselves outside, in the northern Swedish forest at night, looking for the way back.

ROLE

Co-creator with Mauro Zannoli.

CONTEXT

Commissioned by Northern Sustainable Futures for iM Konsthall. Developed during an artist residency in Moskosel, Sweden, October 2023.

Returning virtual space to a place

The container is both the physical venue and an image of transit. Its windows initially frame the same landscape seen before putting on the headset. That continuity provides the perceptual anchor from which the installation departs.

The journey returns to its starting point, placing the familiar room in relation to the altered spaces encountered along the way.

*A virtual journey that begins
and ends in the same room.*

First episode: iM Konsthall, Moskosel, Sweden, 2023.

Manuel Eguía / Mauro Zannoli

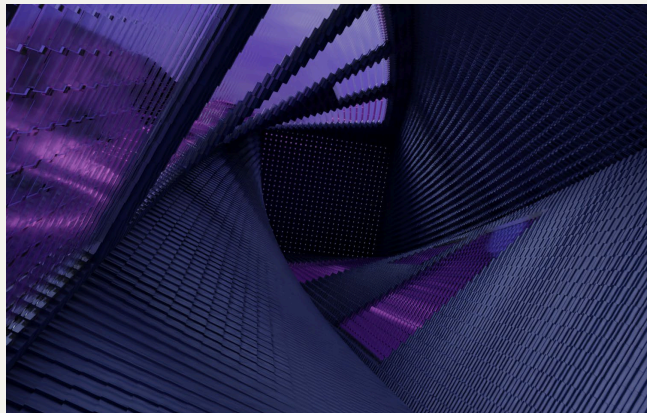
Commission: Northern Sustainable Futures

[Watch the virtual journey ↗](#)

[Northern Sustainable Futures ↗](#)



01 / A familiar room



02 / Spatial transformation



03 / Returning from outside

Proyecto GRAPa

Listening to a voice and the place that answers it

Proyecto GRAPa · Manuel Eguía, Francisco Durante, Damián Payo, Mauro Zannoli



Acoustic heritage · Immersive recording · Documentary

[Watch the copla recording · 360° ↗](#)

[El Anfiteatro · documentary · 11:39 ↗](#)

An enquiry into acoustic heritage through the relationship between copla singing, its communities and the natural spaces in which it is practised.

GRAPa — Grupo de Relevamiento Acústico del Patrimonio — was founded by Manuel Eguía in 2017. We combine acoustic measurement, immersive audiovisual recording and documentary filmmaking to investigate musical practices and their acoustic environments.

La Copla y el Anfiteatro de la Quebrada de las Conchas centres on a natural amphitheatre in the Calchaquí Valleys. With the active participation of coplera Mariana Carrizo and local singers, we recorded the interplay of voices, cajas and rock walls. The acoustic character of the place is part of the musical experience.

ROLE

Founder and member of the research and production team.

CONTEXT

Formed in 2017 through cooperation between UNQ and Universidad Católica de Salta. Fieldwork at the natural amphitheatre of Quebrada de las Conchas, Salta, April 2018. Latin GRAMMY Cultural Foundation research support, 2021.

Three ways of documenting a listening culture



01 / IMMERSIVE RECORDING

Our immersive copla recordings place the listener within the singing circle, preserving the spatial relationships between performers. The work is grounded in the knowledge and participation of the singers, including Mariana Carrizo.

02 / ACOUSTIC SURVEY

We used Ambisonic and binaural impulse-response measurements to capture how the amphitheatre responds to sound. These recordings allow aspects of its acoustic character to be recreated through loudspeakers or headphones.

03 / DOCUMENTARY

The documentary *El Anfiteatro* (2018) follows our April fieldwork and brings the acoustic survey together with the recording of coplas. The three strands document musical practice alongside the space in which it takes place.

Founder: Manuel Eguía

Team: Francisco Durante, Damián Payo, Mauro Zannoli and Manuel Eguía.

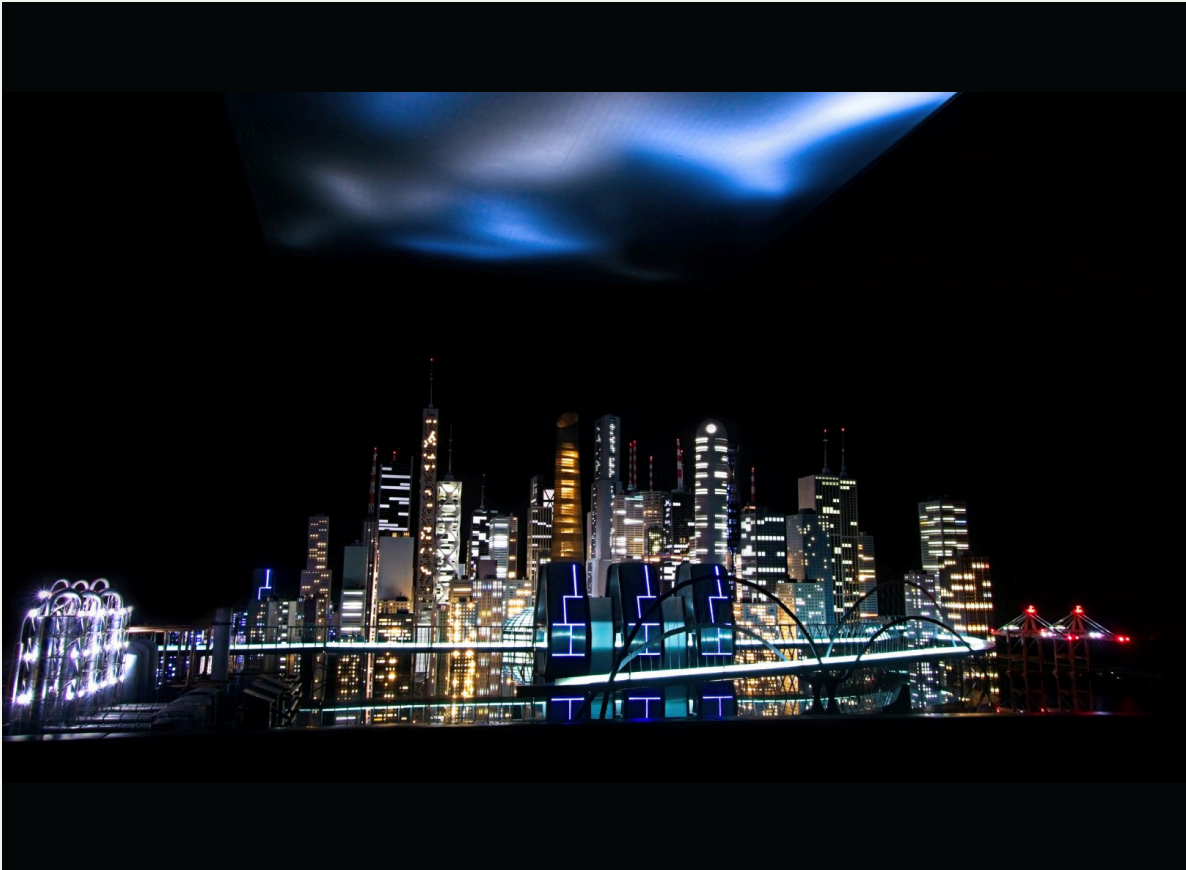


La Copla y el Anfiteatro de la Quebrada de las Conchas · Salta, Argentina
GRAPa, with Mariana Carrizo and local copleras and copleros.

Biocenosis

A city as an interdependent system

Sigismond de Vajay & Kevin Lesquenner + LAPSo



Interactive installation · Sculpture, light and sound

[Watch the installation ↗](#)

[Explore the VR version ↗](#)

An immersive city model stages the interdependence of human, urban and non-human life under environmental pressure.

A model of 430 buildings represents a submerged urban landscape. Sculpture, lighting and interactive systems respond to visitors' presence, changing the behaviour of the fictional city.

The installation was developed by Sigismond de Vajay in collaboration with Kevin Lesquenner and LAPSo, using systems inspired by biological processes. As the LAPSo team, we contributed to the work collectively.

ROLE

Collaborating artist and researcher within LAPSo. Biennial team: Manuel Eguía, Lucas González, Pablo Riera and Sol Gómez.

CONTEXT

13th Mercosul Biennial, Porto Alegre, Brazil, 15 September–20 November 2022.

The laboratory, the classroom, the work

At LAPSo, researchers and artists work together on acoustic experimentation, sound spatialization, instrument design, auditory perception and virtual environments. We develop research and artistic projects within this shared laboratory.

Eguía's teaching includes artist training and project supervision in music, performance and the visual arts. He also supervises doctoral work across science, art and technology and contributes to international workshops involving participants from different disciplines.

Workshops at Performing Arts Forum

L.I.F.E. project, Performing Arts Forum, Saint-Erme, France. Initiated by Gabriel Catren.

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| 2022 | Deeply Organized Chaos
Workshop leader. Chaos and complexity explored through accessible concepts, fractals, sound experiments and embodied experience. 7–13 July. |
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| 2023 | Hot Identities and Vicious Abstractions
Assisted Gabriel Catren, with Jonathan Lorand, in a workshop on the foundations of mathematics. 21–29 October. |
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| 2025 | The Infinite, the Infinitesimal and the Imaginary
Assisted Gabriel Catren, with Jonathan Lorand and Emily Roff. An exploration of calculus, infinity and mathematical thought. 20–28 October. |
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Selected collaborations

Sound design and technological development in dialogue with the authors of these works.

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| 2023–2024 | Beehive / Colmena
Jorge Macchi
Multichannel sound realization. What the Night Tells the Day, PAC, Milan. |
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| 2023 | Glockenbuch IV (Spectre Maria dei Carmini)
Marcus Schmickler, in collaboration with Julian Rohrer
Multichannel sound collaboration. Sound Microscopies programme, Biennale Musica, Venice. |
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| 2021 | Reliquias
Jorge Caterbetti
Technical collaboration on the installation dedicated to Julio López. Centro Cultural Kirchner, Buenos Aires. |
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| 2020 | La Cathédrale engloutie
Jorge Macchi
Development of the sound installation, with UNQ support. Musée cantonal des Beaux-Arts, Lausanne. 10 September–22 November. |
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| 2019 | Huella Sonora / Alfombra Sonora
Jorge Caterbetti
Technical collaboration on an interactive sound installation within Green Carpet. Centro Cultural de la Ciencia, Buenos Aires. |
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Selected CV

Selected work in research, teaching and artistic practice.

Research & teaching

2026–	Director, ArCiTeS Instituto de Investigaciones en Arte, Ciencia, Tecnología y Sociedad, <u>Universidad Nacional de Quilmes</u> .
2019–	Full Professor Escuela Universitaria de Artes, <u>Universidad Nacional de Quilmes</u> .
2017	Founder, Proyecto GRAPa Grupo de Relevamiento Acústico del Patrimonio.
2007	Founder, <u>LAPSo</u> Laboratory of Acoustics and Sound Perception, <u>Universidad Nacional de Quilmes</u> .
	CONICET researcher Physics, nonlinear dynamics, neuroscience and acoustics.
Director of Perspectiva Acústica, Etapa II (2025–2029), a UNQ research programme comprising seven projects.	

Education

2002	PhD in Physics Universidad de Buenos Aires.
1998	Licenciatura in Physics Universidad de Buenos Aires.

Exhibitions, residencies & support

2024	Sonic Crystal Room Festival de Nueva Ópera, opening programme, cheLA, Buenos Aires. With Oscar Edelstein.
2023	Transit · Artist residency & commission Northern Sustainable Futures, Moskosel, Sweden. With Mauro Zannoli, for iM Konsthall.
2022	Biocenosis · 13th Mercosul Biennial Porto Alegre, Brazil. Sigismond de Vajay & Kevin Lesquenner + <u>LAPSo</u> .
2021	Latin GRAMMY Cultural Foundation Research Grant With Proyecto GRAPa: recording canto con caja in its original acoustic environments.
2019	Sonic Crystal Room · First presentation Auditorio Nicolás Casullo, <u>Universidad Nacional de Quilmes</u> , Bernal.
2019	IRIS · Sonic Crystal Room programme Scientific advice and development with Leonardo Salzano. UNQ, Bernal.
2011	The Wind Chimes / Lluvia de Arañas sobre el Riachuelo Instrument design, construction and performance with Buenos Aires Sonora and <u>LAPSo</u> . La Noche de los Museos, Buenos Aires.

Selected publications

Selected studies on sonic crystals, acoustic time reversal, spatial perception, musical timbre and sound synthesis.

Current research: acoustic time reversal

Current experiments extend research on acoustic time reversal to pulses travelling through steel sheets. The work examines how bending the surface changes sound focusing, timbre and radiation, and how these changes relate to gesture and spatial listening.

meagua@unq.edu.ar

[meagua.github.io/dossier](https://github.com/manueleguia/dossier)

vimeo.com/meagua

youtube.com/@manueleguia2915

Work and image credits appear on the relevant project pages.

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| 2026 | The influence of visual context on auditory distance perception: A comparison between real and virtual visual environments
Cerviño, Lombera, Picada, Eguía & Vergara · Applied Acoustics 250, 111334 |
| 2026 | Real-time sound synthesis from normal forms of dynamical systems
Ortega Brook, Ruiz & Eguía · Journal of New Music Research · Online publication 2026 |
| 2025 | Encrypted acoustical communication system using time reversal through a tunable sonic crystal
Gómez, Spiouzas & Eguía · The Journal of the Acoustical Society of America 158(3), 2100–2110 |
| 2022 | IRIS: A tunable sound sculpture based on acoustic periodic composites for musical performance
Salzano & Eguía · Journal of New Music Research 51, 27–39 |
| 2021 | Time reversal focusing in the audible range using a tunable sonic crystal
Gómez, Spiouzas & Eguía · The Journal of the Acoustical Society of America 149(6), 4024–4035 |
| 2017 | Auditory environmental context affects visual distance perception
Etchemendy, Abregú, Calcagno, Eguía, Vechiatti, Iasi & Vergara · Scientific Reports 7, 7189 |
| 2015 | An auditory illusion of proximity of the source induced by sonic crystals
Spiouzas, Etchemendy, Vergara, Calcagno & Eguía · PLOS ONE 10, e0133271 |
| 2014 | A comparative study of saxophone multiphonics: Musical, psychophysical and spectral analysis
Riera, Proscia & Eguía · Journal of New Music Research 43(2), 202–213 |